23/03/2024

Code-A

Corporate Office : Aakash Tower, 8, Pusa Road, New Delhi-110005,
Ph.011-47623456

MM : 720 NCERT Booster Test Series-RM(P1)2324-T07A **Time : 200 Min.**

Topics Covered:
Physics: Ray Optics & Optical Instruments, Wave optics, Dual Nature of Matter and Radiation; Atoms, Nuclei; Semiconductors
Chemistry: The *p*-Block Elements, Biomolecules
Botany: Organisms and Populations, Ecosystem, Biodiversity and Conservation
Zoology: Biotechnology-Principles and Processes, Biotechnology and its Applications

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

SECTION-A

1. The magnifying power of a telescope is 6 when it is adjusted for parallel final rays and the distance between the objective and the eyepiece is found to be 35 cm. The focal length of objective and eye lenses are respectively
(1) 30 cm, 5 cm
(2) 30 cm, 10 cm
(3) 15 cm, 15 cm
(4) 25 cm, 10 cm

2. When light travels from an optically rarer medium to an optically denser medium, which of the following quantities remains unchanged?
(1) Frequency
(2) Speed
(3) Amplitude
(4) Wavelength

3. A equiconvex lens of glass of $\mu = 1.5$ has radius of curvature is 40 cm. The focal length of the lens will be
(1) 10 cm
(2) 40 cm
(3) 20 cm
(4) 15 cm

4. Which of the following will be due to total internal reflection?
(1) Propagation of light in optical fibre
(2) Mirage
(3) Brilliance of diamond
(4) All of these

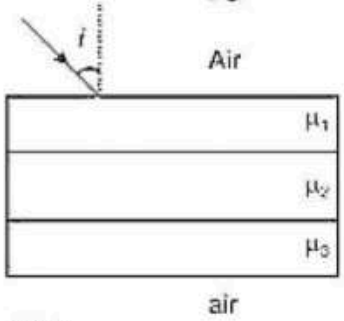
5. A convex lens of focal length 12 cm is made of glass of refractive index $\frac{3}{2}$. What will be its focal length, when it is immersed in liquid of refractive index $\frac{5}{4}$?

(1) 6 cm
(2) 12 cm
(3) 24 cm
(4) 30 cm

6. A point object is kept at 30 cm from convex lens of focal length f = 20 cm. A convex mirror is kept at 20 cm from the lens so that virtual object for convex mirror lies at its centre of curvature. The focal length of the mirror is

(1) 10 cm
(2) 15 cm
(3) 20 cm
(4) 40 cm

7. It is given that $\mu_1 > \mu_2 > \mu_3$. Find angle made by the ray of light with the normal in medium of refractive index (μ_3).



(1) i
(2) $\sin^{-1}\left(\frac{\sin i}{\mu_1}\right)$
(3) $\sin^{-1}\left(\frac{\sin i}{\mu_2}\right)$
(4) $\sin^{-1}\left(\frac{\sin i}{\mu_3}\right)$

8. A thin prism of 4° is made of refractive index $\frac{3}{2}$. The deviation caused by it for small angle of incidence is

(1) 1°
(2) 2°
(3) 6°
(4) 3°

9. Which of the following is not essential condition for two sources of light in YDSE to produce a sustained interference?

(1) Equal wavelength
(2) Equal intensity
(3) Constant phase relation
(4) Equal frequency

10. In Young's double slit experiment, 12 fringes are obtained in a certain segment of the screen when light of wavelength 600 nm is used. If the wavelength of light is changed to 400 nm, the number of fringes observed in the same segment of screen will be

(1) 12
(2) 18
(3) 24
(4) 30

11. For constructive interference at a point on screen in YDSE, which of the following may be the path difference between the light waves coming from the two slits? (λ is the wavelength of light used)

(1) 2λ
(2) $\frac{\lambda}{2}$
(3) $\frac{3\lambda}{2}$
(4) $\frac{7\lambda}{2}$

12. The momentum of an electron having a wavelength of 3 Å is

(1) $6.62 \times 10^{-24}\text{kgms}^{-1}$
(2) $2.21 \times 10^{-34}\text{kgms}^{-1}$
(3) $2.21 \times 10^{-14}\text{kgms}^{-1}$
(4) $2.21 \times 10^{-24}\text{kgms}^{-1}$

13. A monochromatic light of frequency 3×10^{14} Hz is produced by a source of power 3×10^{-5} W. The number of photons emitted per second are

(1) 1.5×10^{16}
(2) 1.5×10^{14}
(3) 1.5×10^{15}
(4) 1.5×10^{17}

14. A light of certain wavelength strikes two metal surfaces with work function ϕ_1 and ϕ_2 respectively (where $\phi_1 > \phi_2$). If k_1 and k_2 are maximum kinetic energies of ejected electrons from these metal surfaces, then

(1) $k_1 > k_2$
(2) $k_2 > k_1$
(3) $k_1 = k_2$
(4) k_1, k_2 can't be compared

15. A proton and an alpha particle are accelerated under same potential difference. The ratio of de-Broglie wavelengths of proton and alpha particle is

(1) $3\sqrt{3}$

(2) $2\sqrt{3}$

(3) $\sqrt{2}$

(4) $2\sqrt{2}$

16. A surface completely absorbs the radiation of intensity *I* falling on it. The radiation pressure is given as (c : speed of light)??

(1) $\frac{I}{c}$

(2) $\frac{2I}{c}$

(3) *Ic*

(4) 2*Ic*

17. If the kinetic energy of a free electron quadruples. Its de-Broglie wavelength change by the factor

(1) 2

(2) $\sqrt{2}$

(3) $\frac{1}{\sqrt{2}}$

(4) $\frac{1}{2}$

18. **Assertion (A):** In photoelectric effect increase in intensity of light increases the photocurrent.
Reason (R): At stopping potential photocurrent is zero.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

19. **Assertion (A):** Microscope magnifies the image.
Reason (R): Telescope resolves the image.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

20. The angular momentum of an electron in hydrogen atom is $\frac{4h}{2\pi}$. The kinetic energy of this electron is

(1) 1.51 eV

(2) 0.85 eV

(3) 1.70 eV

(4) 3.4 eV

21. According to Bohr's quantization rule, the angular momentum of electron in a hydrogen atom in n^{th} orbit is given by

(1) $\frac{nh}{2\pi}$

(2) $\frac{n^2h}{2\pi}$

(3) $\frac{nh}{4\pi}$

(4) $\frac{n^2h}{4\pi}$

22. In hydrogen atom, the transition (among the following) for which minimum energy is emitted is (symbols have their usual meanings)

(1) $n_1 = 2, n_2 = 3$

(2) $n_1 = 2, n_2 = 4$

(3) $n_1 = 3, n_2 = 4$

(4) $n_1 = 4, n_2 = 5$

23. The ratio of maximum to minimum possible radiation energy in Bohr's hypothetical hydrogen atom (In Lyman series) is equal to

(1) 2

(2) $\frac{1}{4}$

(3) $\frac{4}{3}$

(4) $\frac{3}{2}$

24. Hydrogen atoms are excited from ground state to another state with principal quantum number equal to 4. Then the number of spectral lines in the emission spectra will be

(1) 2

(2) 3

(3) 5

(4) 6

25. If the value of Rydberg constant is $10^7\ m^{-1}$, the wave number of the last line of Lyman series in hydrogen spectrum will be

(1) $0.75 \times 10^7\ m^{-1}$

(2) $0.25 \times 10^7\ m^{-1}$

(3) $10^7\ m^{-1}$

(4) $0.5 \times 10^7\ m^{-1}$

26. When aluminium is bombarded with fast moving neutron, it changes into sodium with emission of particle *X* according to equation
$^{27}_{13}\mathrm{Al} + {}^{1}_{0}n \rightarrow {}^{24}_{11}\mathrm{Na} + X$
The particle *X* is
(1) Electron
(2) Proton
(3) Alpha particle
(4) Positron

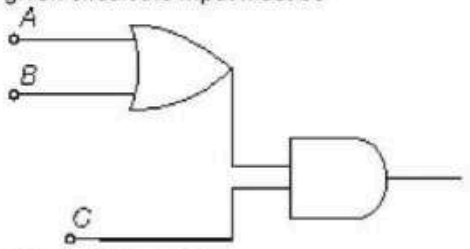
27. The controlling rod in a nuclear reactor is made of
(1) Uranium
(2) Cadmium
(3) Graphite
(4) Plutonium

28. The binding energy of deuteron $\left({}^{2}_{1}\mathrm{H}\right)$ is 1.112 MeV per nucleon and an α-particle $\left({}^{4}_{2}\mathrm{He}\right)$ has a binding energy of 7.047 MeV per nucleon. Then in the fusion reaction ${}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + \mathrm{Q}$. The energy *Q* released is
(1) 1 MeV
(2) 11.9 MeV
(3) 23.8 MeV
(4) 931 MeV

29. The atomic mass of $^{234}_{90}\mathrm{Th}$ is 234.04359 u and that of $^{234}_{91}\mathrm{Pa}$ is 234.04330 u. The energy released in the reaction $^{234}_{90}\mathrm{Th} \longrightarrow {}^{234}_{91}\mathrm{Pa} + {}^{0}_{-1}\beta$ will be
(1) 0.27 MeV
(2) 0.54 MeV
(3) 0.12 MeV
(4) 0.74 MeV

30. The ratio of radius of nuclei $_{13}Al^{27}$ and $_{52}X^{A}$ is 3 : 5. The value of *A* is
(1) 81
(2) 73
(3) 65
(4) 125

31. Among the following to get an output 1 from given circuit the input must be



(1) $A = 0, B = 1, C = 0$
(2) $A = 1, B = 0, C = 0$
(3) $A = 1, B = 0, C = 1$
(4) $A = 1, B = 1, C = 0$

32. In a extrinsic semiconductor, the impurity level is just below conduction band. The semiconductor is
(1) p-type
(2) n-type
(3) Intrinsic
(4) May be p-type or n-type

33. An LED operates (properly) under the condition of
(1) Reverse bias
(2) Forward bias
(3) Both in forward and reverse bias
(4) No biasing

34. When a semiconductor is heated its resistance
(1) Decreases
(2) Increases
(3) Remain unchanged
(4) Nothing is defined

35. An alpha particle having energy $\frac{1}{2}mv^2$ bombards a heavy nuclear target of charge *ze*. The distance of closest approach for the alpha nucleus will be proportional to
(1) z
(2) $\frac{1}{m}$
(3) $\frac{1}{v^4}$
(4) Both (1) and (2)

SECTION-B

36. The refracting angle of prism is 60° and the refractive index of material of prism is $\sqrt{3}$. The angle of minimum deviation is

(1) 180°

(2) 45°

(3) 30°

(4) 60°

37. If the velocity of light in vacuum be c, then velocity of light in a medium of refractive index $\frac{4}{3}$ is

(1) $\frac{3}{4}c$

(2) $\frac{4}{3}c$

(3) $\frac{1}{3}c$

(4) c

38. Young's double slit experiment is first performed in air and then in a medium other than air. It is found that 8th bright fringe in the medium lies where 5th dark fringe lies in air. The refractive index of the medium is nearly

(1) 1.25

(2) 1.59

(3) 1.69

(4) 1.78

39. If the amplitude of two waves from coherent sources producing interference is 1 : 2, the ratio of intensities at maxima and minima is

(1) 9 : 1

(2) 25 : 16

(3) 25 : 9

(4) 16 : 1

40. When interference of light takes place:

(1) Energy is created in the region of maximum intensity

(2) Energy is destroyed in the region of maximum intensity

(3) Conservation of energy holds good and energy is redistributed

(4) Conservation of energy does not hold good and energy is redistributed

41. Light of frequency 1.5 times the threshold frequency is incident on a photosensitive material. If the frequency is halved and intensity is doubled, the photo current becomes

(1) Quadrupled

(2) Doubled

(3) Halved

(4) Zero

42. Which among the following shows the particle nature of light?

(1) Interference

(2) Polarisation

(3) Photoelectric effect

(4) Diffraction

43. An electron is accelerated from rest by 100 V de-Broglie wavelength of the electron is

(1) 1.227 Å

(2) 12.27 Å

(3) 122.7 Å

(4) 0.1227 Å

44. Potential energy of 3rd Bohr orbit of H-atom should be

(1) – 3.4 eV

(2) – 3.02 eV

(3) – 1.51 eV

(4) – 13.6 eV

45. Rutherford's experiments suggested that the size of the nucleus to be around

(1) $(10^{-15}$ to $10^{-14})$ cm

(2) $(10^{-13}$ to $10^{-12})$ cm

(3) $(10^{-14}$ to $10^{-12})$ m

(4) $(10^{-10}$ to $10^{-9})$ m

46. The first line in the Lyman series has wavelength λ. The wavelength of the first line in Balmer series is

(1) $\frac{2}{9}\lambda$

(2) $\frac{9}{2}\lambda$

(3) $\frac{5}{27}\lambda$

(4) $\frac{27}{5}\lambda$

47. In a fission process nucleus A divided into two nuclei B and C. Their binding energies are E_a , E_b and E_c respectively. Then

(1) $E_b + E_c = E_a$

(2) $E_b + E_c > E_a$

(3) $E_b + E_c < E_a$

(4) $E_b - E_c = E_a$

48. The energy released by fission of one uranium atom is 200 MeV. The number of fissions per second required to produce 6.4 W power will be

(1) 10^{11}

(2) 2×10^{11}

(3) 10^{10}

(4) 2×10^{10}

49. In an n-type semiconductor if n_1 is the number of conduction electrons and n_2 is the number of holes, then,

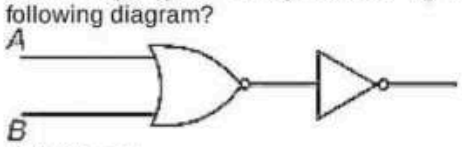
(1) $n_1 >> n_2$

(2) $n_1 << n_2$

(3) $n_1 = n_2$

(4) $n_1 \neq 0, n_2 = 0$

50. Which logic gate is represented by the following diagram?



(1) OR gate

(2) NOR gate

(3) NOT gate

(4) NAND gate

CHEMISTRY

SECTION-A

51. In vapour state, sulphur (S_2) shows paramagnetic behaviour due to

(1) Presence of one unpaired electron in the antibonding σ^* orbitals

(2) Presence of two unpaired electrons in the bonding π orbitals

(3) Presence of one unpaired electron in the antibonding π^* orbitals.

(4) Presence of two unpaired electrons in the antibonding π^* orbitals.

52. Deacon's process is

(1) $MnO_2 + 4\ HCl \rightarrow MnCl_2 + Cl_2 + 2\ H_2O$

(2) $4\ NaCl + MnO_2 + 4\ H_2SO_4 \rightarrow MnCl_2 + 4\ NaHSO_4 + 2\ H_2O + Cl_2$

(3) $2\ KMnO_4 + 16\ HCl \rightarrow 2\ KCl + 2\ MnCl_2 + 5\ Cl_2$

(4) $4HCl + O_2 \xrightarrow{CuCl_2} 2Cl_2 + 2H_2O$

53. The gas which is released by thermal decomposition of $(NH_4)_2Cr_2O_7$ is

(1) O_2

(2) N_2

(3) NO

(4) NH_3

54. Number of P–H bonds in orthophosphoric acid is

(1) Zero

(2) 1

(3) 2

(4) 3

55. Most basic hydride of group 15 elements is

(1) NH_3

(2) PH_3

(3) AsH_3

(4) SbH_3

56. Which is correctly matched?

(1) H_2SO_4 – Haber's process

(2) HNO_3 – Contact process

(3) Cl_2 – Deacon's process

(4) HCl – Ostwald process

57. Cl_2 on reaction with cold and dilute NaOH gives NaCl and

(1) NaOCl

(2) $NaClO_3$

(3) $NaClO_4$

(4) $NaClO_2$

58. Correct order of decreasing pK_a of HCl, HBr, HI and HF is

(1) HCl > HF > HBr > HI

(2) HF > HCl > HBr > HI

(3) HI > HBr > HCl > HF

(4) HI > HBr > HF > HCl

59. The neutral oxide of Nitrogen is

(1) N_2O_5

(2) N_2O

(3) N_2O_4

(4) NO_2

60. Hydride of group-16 having lowest boiling point is

(1) H_2O

(2) H_2S

(3) H_2Se

(4) H_2Te

61. Which ion gives brown ring test?

(1) Cl^-

(2) NO_3^-

(3) SO_3^{2-}

(4) $\overset{\ominus}{N}H_2$

62. Cationic species formed on reaction of XeF_2 with PF_5 is

(1) $[XeF_3]^+$

(2) $[XeF]^+$

(3) $[XeF_5]^+$

(4) $[XeF_4]^+$

63. When Cu reacts with $AgNO_3$ solution, the reaction takes place is

(1) Oxidation of Cu

(2) Reduction of Cu

(3) Oxidation of Ag

(4) Reduction of NO_3^-

64. Which of the following bases is not present in DNA?

(1) Guanine

(2) Adenine

(3) Uracil

(4) Cytosine

65. Fat Soluble vitamin among the following is

(1) Thiamine

(2) Riboflavin

(3) Vitamin D

(4) Vitamin-C

66. Glucose reacts with hydroxylamine to form an oxime. This reaction confirms

(1) The presence of – CN group

(2) The presence of primary OH group

(3) The presence of –COOH group

(4) The presence of CHO group or  group

67. Which of the following can't form Zwitter ion

(1) Glycine

(2) Valine

(3) Lysine

(4) Aniline

68. Which among the following is an optically inactive amino acid?

(1) Proline

(2) Lysine

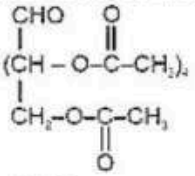
(3) Glycine

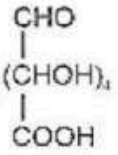
(4) Alanine

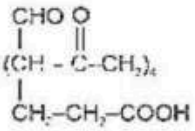
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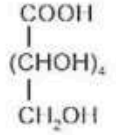


The product of the above reaction would be

(1) 

(2) 

(3) 

(4) 

70. Which among the following is an essential amino acid?
(1) Histidine
(2) Alanine
(3) Proline
(4) Serine

71. Which hormone is released in response to the rapid rise in blood glucose level?
(1) Glucagon
(2) Insulin
(3) Thyroxine
(4) Epinephrine

72. Non-reducing carbohydrate among the following is
(1) Sucrose
(2) Glucose
(3) Lactose
(4) Maltose

73. Lactose is composed of
(1) β-D-Glucose and β-D-Fructose
(2) β-D-Galactose and β-D-Glucose
(3) β-D-Fructose and α-D-Glucose
(4) β-D-Galactose and α-D-Glucose

74. Enzymatic reaction of zymase enzyme is
(1) To convert sucrose into glucose and fructose
(2) To convert glucose into ethyl alcohol and carbon dioxide
(3) To convert urea into ammonia and carbon dioxide
(4) To convert proteins into amino acids

75. **Statement-I :** Amylose is water insoluble component of starch.
Statement-II : Amylopectin is water soluble component of starch.
In the light of above statements choose the correct option among the following.
(1) Both statement-I and statement-II are incorrect
(2) Both statement-I and statement-II are correct
(3) Statement-I is incorrect but statement-II is correct
(4) Statement-I is correct but statement-II is incorrect

76. Incorrect statement among the following is
(1) Proline is a non-essential amino acid
(2) Glycine is an optically inactive molecule
(3) Myosin is fibrous protein
(4) Insulin has fibre-like structure

77. According to inert pair effect, Pb shows its more stable oxidation state as
(1) + 1
(2) + 2
(3) + 3
(4) + 4

78. A : H_3PO_2 is a stronger reducing agent than H_3PO_4.
R : H_3PO_2 has no OH group.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

79. A : Graphite is used to make electrodes of electric furnaces.
R : Graphite is good conductor of electricity.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

80. Which compound on hydrolysis gives cross-linked polymer?
(1) $(CH_3)_3SiCl$
(2) $(CH_3)_2SiCl_2$
(3) CH_3SiCl_3
(4) $(Ph)_2SiCl_2$

81. Which of the following is good conductor of electricity?
(1) Solid NaCl
(2) CO_2
(3) Graphite
(4) Benzene

82. A : C, Si and Ge are not affected by water.
R : Tin decomposes when reacts with steam to form dioxide and dihydrogen gas.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

83. α-D-glucose and β-D-glucose are related as
(1) Anomers
(2) Enantiomers
(3) Conformers
(4) Geometrical isomers

84. A : SO_2 is anhydride of sulphurous acid.
R : When SO_2 is passed through lime water, it becomes milky.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

85. A : F_2 is stronger oxidising agent than Cl_2.
R : Negative electron gain enthalpy of chlorine is more than fluorine.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

SECTION-B

86. Which compound of xenon is of square planar structure?
(1) XeF_2
(2) XeO_3
(3) XeF_4
(4) $XeOF_4$

87. The key step in the manufacture of H_2SO_4 by contact process is
(1) Burning of sulphur to generate SO_2
(2) Conversion of SO_2 to SO_3 in the presence of catalyst
(3) Absorption of SO_3 in H_2SO_4
(4) Dilution of oleum ($H_2S_2O_7$)

88. Cl_2 shows bleaching action due to the formation of
(1) Cl^-
(2) OCl^-
(3) OH^-
(4) All of these

89. The solution used to dissolve noble metals is prepared by mixing
(1) 2 parts HCl + 1 part H_2SO_4
(2) 3 parts HCl + 1 part HNO_3
(3) 3 parts HNO_3 + 1 part HCl
(4) 2 parts H_2SO_4 + 1 part HCl

90. XeF_2 is hydrolysed to produce
(1) Xe
(2) XeO_3
(3) XeO_2F_2
(4) $XeOF_4$

91. Pyridoxine is also known as
(1) Vitamin B_6
(2) Vitamin C
(3) Vitamin B_1
(4) Vitamin B_2

92. Correct statement regarding DNA and RNA is
(1) Sugar component in DNA is α-D-ribose and that in RNA is β-D-2-deoxyribose
(2) Sugar component in DNA is β-D-2-deoxyribose and that in RNA is β-D-ribose
(3) Sugar component in DNA is α-D-2-deoxyribose and that in RNA is α-D-2-arabinose
(4) Sugar component in DNA is β-D-arabinose and that in RNA is α-D-arabinose

93. Which of the following carbohydrates cannot be hydrolysed further to give simpler unit of polyhydroxy aldehyde or ketone?
(1) Sucrose
(2) Glucose
(3) Cellulose
(4) Starch

94. Glucose on reaction with Br_2 water forms
(1) n-Hexane
(2) Gluconic acid
(3) Saccharic acid
(4) Adipic acid

95. Glucose on prolonged heating with HI gives
(1) Gluconic acid
(2) n-Hexane
(3) Saccharic acid
(4) Hexanoic acid

96. Which of the following compounds do not undergo hydrolysis?
(1) $SiCl_4$
(2) CCl_4
(3) $TlCl_3$
(4) BCl_3

97. H_3PO_3 on heating gives
(1) H_3PO_4
(2) PH_3
(3) H_3PO_2
(4) Both (1) and (2)

98. The most powerful reducing agent is
(1) H_3PO_3
(2) H_3PO_4
(3) $H_4P_2O_7$
(4) $H_4P_2O_6$

99. Bonds present in NH_4Cl is/are
(1) Covalent and coordinate only
(2) Covalent, ionic and coordinate
(3) Only covalent
(4) Only ionic

100. A mixed anhydride is
(1) N_2O_5
(2) NO_2
(3) N_2O_3
(4) N_2O

BOTANY

SECTION-A

101. Monarch butterfly shows
(1) Presence of spines
(2) Chemical defence
(3) Mimicry
(4) Defence by sharp appendages

102. Which one is **odd** w.r.t. population attributes?
(1) Birth rate
(2) Sex ratio
(3) Death
(4) Death rate

103. Competitive exclusion principle was given by
(1) J. Grinnel
(2) Gause
(3) E.P. Odum
(4) Ernst Haeckel

104. The rate of formation of new organic matter by consumers is
(1) Primary productivity
(2) Net primary productivity
(3) Gross primary productivity
(4) Secondary productivity

105. Humus is **not**
(1) A dark coloured compound
(2) An amorphous substance
(3) Easily degraded by microbes
(4) More or less decomposed organic matter

106. Which one is **not** a key functional aspect of an ecosystem?
(1) Decomposition
(2) Energy flow
(3) Species composition
(4) Nutrient cycling

107. What percentage of PAR is captured by plants?
(1) More than 50%
(2) 50%
(3) 20-40%
(4) 2-10%

108. Primary productivity does not depend upon
(1) Plant species inhabiting a particular area
(2) Photosynthetic capacity of plants
(3) Biomass of parasites
(4) Availability of nutrients

109. Each trophic level has a certain mass of living material at a particular time called as the
(1) Standing crop
(2) Standing state
(3) Primary productivity
(4) Gross productivity

110. Out of the total quantity of global carbon, 71% is found in the
(1) Oceans
(2) Agroecosystems
(3) Grasslands
(4) Forests

111. The historical 'Convention on Biological Diversity' known as 'Earth Summit' was held in
(1) South Africa in 1999
(2) Montreal in 1987
(3) Rio-de-Janeiro in 1992
(4) Johannesburg in 1992

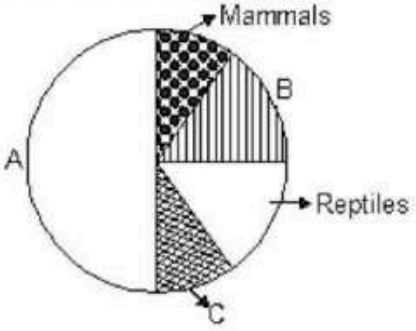
112. Robert May placed the global species diversity at about
(1) 20 million
(2) 50 million
(3) 1.5 million
(4) 7 million

113. The organization which publishes the Red List of species is
(1) IARI
(2) IUCN
(3) KVIC
(4) ICAR

114. "Rivet popper hypothesis" was used to explain the effect of reduction in biodiversity on ecosystem by
(1) David Tilman
(2) Paul Ehrlich
(3) Alexander von Humboldt
(4) Tansley

115. One of the reasons of the richness of biodiversity in tropics is
(1) Frequent glaciations in the past
(2) Being more seasonal and less constant
(3) Being more productive for supporting wide range of species
(4) High temperature and humidity but receiving less solar radiations

116. Observe the given pie chart representing biodiversity of different vertebrates. Choose the correct option for (A – C)



	A	B	C
(1)	Fishes	Birds	Amphibians
(2)	Fishes	Molluscs	Amphibians
(3)	Insects	Birds	Fishes
(4)	Insects	Crustaceans	Molluscs

(1) (1)
(2) (2)
(3) (3)
(4) (4)

117. State true (T) or false (F) and choose the **correct** option

A. India has a greater ecosystem diversity than a Scandinavian country like Norway.

B. Tilman showed that in his experiments, increased diversity contributed to higher productivity.

C. Tropical amazonian rain forest is home to 40000 species of plants, 3000 species of birds and 1300 species of reptiles.

	A	B	C
(1)	T	T	T
(2)	F	T	T
(3)	T	F	F
(4)	T	T	F

(1) (1)
(2) (2)
(3) (3)
(4) (4)

118. Read the following statements and choose the **correct** option.

Assertion (A) : GFC is major conduit for energy flow in aquatic ecosystem.

Reason (R) : DFC may be connected with GFC at some levels.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

119. Choose the **correct** sequence of successional communities in hydrarch succession.

(1) Submerged plants → Phytoplankton → Reed swamp → Forest
(2) Phytoplankton → Submerged plant → Submerged free floating plants → Reed swamp
(3) Lichens → Bryophytes → Herbs → Trees
(4) Phytoplankton → Herbs → Bryophytes → Trees

120. Choose the option having **correct** set of statements.

(a) Decomposition rate is slower if detritus is rich in lignin and chitin.

(b) Temperature and soil moisture are the most important climatic factors that regulate decomposition.

(c) Warm and moist environment do not favour decomposition.

(d) Low temperature and anaerobiosis favour decomposition.

(1) (a) and (c)
(2) (b) and (d)
(3) (c) and (d)
(4) (a) and (b)

121. Which of the following is the major and important difference between carbon and phosphorous cycle?

(1) Atmospheric inputs of phosphorus through rainfall are much smaller than carbon inputs
(2) Gaseous exchanges of carbon between organisms and environment are negligible
(3) Unlike phosphorus cycle, there is no respiratory release of carbon into atmosphere
(4) Phosphorus constitutes 49 percent of dry weight of organisms and is next only to water

122. Regarding the limitations of ecological pyramids, which of the following statement(s) is/are **incorrect**?

a. It assumes a food chain as well as food web.

b. Saprophytes are given a place in the pyramids but microbes do not

c. It does not take into account the insectivorous animals

(1) a only
(2) All a, b and c
(3) a and b only
(4) b and c only

123. Find out the **correct** match:

	Ecosystem service	**Cost**
(a)	Nutrient cycling	> 10 %
(b)	Climate regulation	About 6 %
(c)	Soil formation	20 %
(d)	Habitat for wildlife	12 %

(1) (a)
(2) (b)
(3) (c)
(4) (d)

124. Which of the following pair(s) are **correctly** matched?
a. Amensalism – A species promotes the growth of the other species by secreting chemicals
b. Predation – A population is inhibited whereas the other population remains unaffected
c. Mutualism – Interaction confers benefits to both interacting species
(1) a, b and c
(2) a and b
(3) Only c
(4) b and c

125. What is **incorrect** for temperature as an abiotic factor of a habitat?
(1) Plays role in distribution of organisms
(2) Most important ecologically relevant factor
(3) Affects the enzyme kinetics but does not affect periodic activities of organisms
(4) Temperature decreases from equator towards the poles on earth

126. Consider the following equation
$\log S = \log C + Z \log A$
In the given species-area relationship equation, C represents
(1) Species richness
(2) Y-intercept
(3) Area
(4) Regression coefficient

127. The lowest level of organization among the following is
(1) Populations
(2) Landscapes
(3) Ecosystems
(4) Communities

128. The annual NPP of ocean is only __A__ and for terrestrial ecosystem, it is __B__.
Select the **correct** option to fill in the blanks.
(1) A – 115 billion tons, B – 55 billion tons
(2) A – 105 billion tons, B – 75 billion tons
(3) A – 55 billion tons, B – 115 billion tons
(4) A – 75 billion tons, B – 105 billion tons

129. In logistic growth a plot of population density (N) in relation to time (t) results in a sigmoid curve, which finally shows an asymptote, when
(1) Intrinsic rate of natural increase is zero
(2) Carrying capacity = Population density
(3) Carrying capacity > Population density
(4) Carrying capacity < Population density

130. Choose the **odd** one for sacred groves.
(1) Khasi and Jaintia hills in Meghalaya
(2) Aravalli hills of Rajasthan
(3) Western Ghat regions of Karnataka and Maharashtra
(4) Pushkar lake in Rajasthan

131. A stage of suspended development found in many zooplanktons is known as
(1) Aestivation
(2) Diapause
(3) Hibernation
(4) Osmoregulation

132. Select the option that correctly identifies P, Q in the equation given below
$N_{t+1} = N_t + [(\text{Natality} + P) - (\text{Mortality} + Q)]$
(1) P = Immigration, Q = Emigration
(2) P = Emigration, Q = Immigration
(3) P = Illuviation, Q = Eluviation
(4) Q = Eluviation, P = Illuviation

133. A population is growing in a habitat which has limited resources. This population will show the sequence of growth phases
(1) Lag phase → Log phase → Crash
(2) Log phase → Lag Phase → Asymptote
(3) Lag phase → Deceleration → Log phase → Acceleration
(4) Lag phase → Acceleration → Deceleration-→ Asymptote

134. Arrange the following biomes in decreasing order of mean annual precipitation.
A. Desert
B. Coniferous forest
C. Grassland
D. Tropical forest
(1) D > B > C > A
(2) A > C > B > D
(3) D > C > B > A
(4) A > B > C > D

135. Select the example of the organism showing behavioural response to cope with variations in their environment
(1) Archaebacteria flourishing in deep sea hydrothermal vents at temperatures above 100°C
(2) Desert lizards basking in sun absorbing heat and moving to shade when ambient temperature increases
(3) Desert plants showing CAM photosynthetic pathway
(4) Fishes thriving in Antarctic waters where the temperature is always below 0ºC

SECTION-B

136. In species-area relationships among very large areas like the entire continents, the values of Z are in the range of

(1) 0.1 to 0.2
(2) 6 to 12
(3) 0.3 to 0.6
(4) 0.6 to 1.2

137. The most important cause of biodiversity loss is

(1) Habitat loss and fragmentation
(2) Alien species invasion
(3) Over-exploitation
(4) Co-extinction

138. Which of the following distribution pattern of birds depicts the latitudinal gradient of biodiversity.

	No. of species of birds			
	1400	1200	105	56
(1)	India	New York	Colombia	Greenland
(2)	Colombia	India	Greenland	New York
(3)	Colombia	India	New York	Greenland
(4)	New York	Greenland	India	Colombia

(1) (1)
(2) (2)
(3) (3)
(4) (4)

139. Read the statements and select the **correct** option.

Statement A: Fruit eating birds are called frugivorous.

Statement B: Current species extinction rates are estimated to be 100 to 1000 times faster than in the pre-human era.

(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

140. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	*Ex-situ* conservation	(i)	Animals (fauna) are protected from all types of exploitations
b.	Sanctuaries	(ii)	Regions of accelerated habitat loss
c.	Hotspots	(iii)	Species confined to a particular area and not found anywhere else
d.	Endemism	(iv)	Zoological parks

(1) a(iv), b(i), c(ii), d(iii)
(2) a(i), b(ii), c(iii), d(iv)
(3) a(iv), b(ii), c(iii), d(i)
(4) a(iv), b(i), c(iii), d(ii)

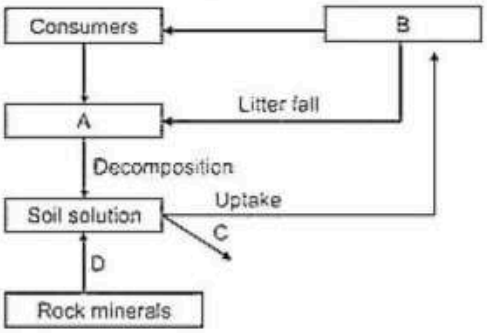
141. The pyramid of biomass in the aquatic ecosystem is usually

(1) Inverted
(2) Upright
(3) Spindle shaped
(4) Either spindle shaped or upright

142. Which of the following is a man-made ecosystem?

(1) Crop-field
(2) Grassland
(3) Desert
(4) Forest

143. Identify the blanks in the given figure and select the **correct** option.



	A	B	C	D
(1)	Detritus	Run off	Producers	Weathering
(2)	Detritus	Producers	Run off	Weathering
(3)	Producers	Detritus	Run off	Weathering
(4)	Weathering	Detritus	Run off	Producers

(1) (1)
(2) (2)
(3) (3)
(4) (4)

144. Consider the following statements, state them as **true (T)** or **false (F)** and choose the **correct** option.
A. The species that invade a bare area are called pioneer species.
B. During xeric succession, the lichens pave way to some very small plants like bryophytes, which are able to take hold in the small amount of soil.
C. The climax community remains stable even if the environment changes.

	A	B	C
(1)	T	F	T
(2)	F	T	T
(3)	T	T	F
(4)	T	T	T

(1) (1)
(2) (2)
(3) (3)
(4) (4)

145. Which among the following can occupy more than one trophic level in ecosystem?
(1) Cow
(2) Sparrow
(3) Phytoplankton
(4) Grasshopper

146. Some snails and fish avoid summer related problems of heat and desiccation by escaping in time and go into
(1) Diapause
(2) Migration
(3) Aestivation
(4) Winter sleep

147. According to Allen's rule, mammals from colder climates generally have _____.
(1) Larger extremities
(2) Shorter ears and longer limbs
(3) Large ears and shorter limbs
(4) Shorter ears and limbs

148. Choose the option for **incorrect** statement(s) for conformers.
A. Aquatic animals change osmotic concentration of body fluids with that of the ambient water osmotic concentration.
B. Body temperature does not change with small fluctuations in environmental temperature.
C. Cannot maintain a constant internal environment.
(1) Only B
(2) Only A
(3) Only A and B
(4) A, B and C

149. Which of the given is **not** an exotic species in India?
(1) *Parthenium*
(2) *Lantana*
(3) *Eicchornia*
(4) Nile perch

150. Read the Assertion (A) and Reason (R) statements and select the **correct** option.
Assertion (A) : *Chathamalus* is competitively superior than *Balanus*.
Reason (R) : *Chathamalus* dominated the intertidal area and replaced *Balanus*.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

ZOOLOGY

SECTION-A

151. Which of the following enzymes is used to digest the cell wall of a fungal cell during isolation of the DNA from it?
(1) Ribonuclease
(2) Chitinase
(3) Protease
(4) Deoxyribonuclease

152. An American company filed a patent for turmeric powder without giving compensation to the authorities of India as since ancient times turmeric is used for its power to cure the wounds. It is an example of
(1) Biosecurity
(2) Biopiracy
(3) Bioethics
(4) Biowar

153. Read the following statements carefully and choose the **correct** option.
Statement X : In the year 1963, the two enzymes responsible for restricting the growth of bacteriophage in *E.coli* were isolated.
Statement Y : Today we know more than 900 restriction enzymes that have been isolated from over 230 strains of bacteria.
(1) Both statements X and Y are correct
(2) Both statements X and Y are incorrect
(3) Only statement X is correct
(4) Only statement Y is correct

154. Biolistics is mostly suitable for
(1) Disarming pathogen vectors
(2) Transformation of plant cells
(3) Transformation of animal cells
(4) Staining of DNA sample

155. A gene has been ligated to the plasmid vector to prepare a recombinant DNA during bacterial transformation. After formation of recombinant DNA, an exonuclease was added to the tube accidentally. How will it affect the next step of the experiment?
(1) Recombinant DNA will be degraded
(2) No effect on the experiment
(3) Alien DNA will not be transferred into host
(4) No product will be formed

156. Read the following statements w.r.t. bioreactors.
a. In bioreactors, 100-1000 litres of cultures can be processed.
b. Optimum growth conditions (temperature, pH, substrate, salts, vitamins, oxygen) are provided for achieving the desired product.
c. In sparged stirred-tank bioreactor, air bubbles dramatically increase the oxygen availability throughout the bioreactor.
Choose the option with **correct** statement(s).
(1) a and b only
(2) c only
(3) a and c only
(4) a, b and c

157. DNA fragments are resolved based on their size in agarose gel electrophoresis through sieving effect. The movement of DNA fragments is based on the property of negative charge on
(1) Sugar moieties
(2) Nitrogenous bases
(3) Phosphate moieties
(4) Nucleoside

158. Gene cloning refers to
(1) Production of mutated harmful genes
(2) Production of multiple identical copies of desired genes
(3) Identification of DNA with desirable genes
(4) Production of large sized DNA with harmful genes

159. A bioreactor does **not** contain
(1) Agitator system
(2) An oxygen delivery system
(3) Foam control system
(4) Quality testing system

160. RNAi which is employed in making tobacco plant resistant to *Meloidegyne incognita* is essentially involved in preventing the process of
(1) Transcription
(2) Translation of the mRNA
(3) Replication of the DNA
(4) Extension of the DNA

161. If a foreign gene is inserted at *Cla*I recognition site in cloning vector pBR322, then the recombinants will be

(1) Sensitive to tetracycline

(2) Resistant to chloramphenicol

(3) Susceptible to both tetracycline and ampicillin

(4) Resistant to both tetracycline and ampicillin

162. While isolating DNA from bacteria, which of the following enzymes is **not** used?

(1) Lysozyme

(2) Deoxyribonuclease

(3) Ribonuclease

(4) Protease

163. If a fragment of dsDNA is amplified 1 billion times, the total number of cycles repeated in PCR is about

(1) 10

(2) 15

(3) 25

(4) 30

164. Which of the following is the recognition sequence of the restriction enzyme isolated from *Escherichia coli*, strain RY 13 ?

(1) 5′ – G A A T T C – 3′
3′ – C T T A A G – 5′

(2) 5′ – G A T A T C – 3′
3′ – C T A T A G – 5′

(3) 5′ – A A G C T T – 3′
3′ – T T C G A A – 5′

(4) 5′ – G G A T C C – 3′
3′ – C C T A G G – 5′

165. In a bioreactor, continuous culture involves maintaining the cells in their physiologically most active _____ phase
Select the **correct** option to fill in the blank.

(1) Declining

(2) Lag

(3) Log

(4) Stationary

166. A bacterium adds methyl groups to its DNA, by a process known as methylation, in order to

(1) Clone its DNA

(2) Activates its gene

(3) Transcribe many genes simultaneously

(4) Protect its DNA from its own restriction enzymes

167. Today, transgenic models have been developed for many human diseases, which includes
I. Rheumatoid arthritis
II. Alzheimer's disease
III. Cancer
IV. Cystic fibrosis
Choose the **correct** option.

(1) I and II only

(2) II and IV only

(3) I, II and IV only

(4) I, II, III and IV

168. Read the statements carefully and choose the correct option
Statement A: The green revolution succeeded in only doubling the food supply.
Statement B: Increased yields of crops was mainly due to the use of improved crop varieties during green revolution.

(1) Both statements are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both statements are incorrect

169. Which of the following techniques does not serve the purpose of early diagnosis of diseases?

(1) Serum analysis test

(2) Recombinant DNA Technology

(3) Polymerase Chain Reaction

(4) Enzyme Linked Immunosorbent Assay

170. Select the correct option to complete the analogy.
Lepidopterans : Tobacco budworm : : Dipterans : _____

(1) Armyworms

(2) Beetles

(3) Mosquitoes

(4) Corn borer

171. Study the following statements w.r.t ADA deficiency and choose the option which **correctly** fills the blanks A and B.
(a). As a first step towards gene therapy, ___A___ from the blood of the patient are grown in a culture outside the body.
(b). A functional ADA cDNA, using a ___B___ vector, is then introduced into these lymphocytes, which are subsequently returned to the patient.

	A	B
(1)	RBCs	Plasmid
(2)	Lymphocytes	Disarmed retrovirus
(3)	Lymphocytes	Bacteriophage
(4)	RBCs	Retroviral

(1) (1)
(2) (2)
(3) (3)
(4) (4)

172. 'Elution' is the technique of
(1) Separating DNA fragments of various sizes through gel matrix under electric field.
(2) Observing bright orange DNA bands in a stained gel under UV radiation.
(3) Extraction of DNA bands from the gel piece used in gel electrophoresis.
(4) Construction of recombinant DNA by joining DNA fragments with cloning vectors.

173. Which of the below statement(s) do(es) **not** belong to the three critical research areas of biotechnology?
a. Providing the best catalyst in the form of improved organism usually a microbe or pure enzyme.
b. Creating optimal conditions through engineering for a catalyst to act.
c. To file a biopatent on the developed novel research.
d. Downstream processing technologies to purify the protein/organic compound.
Select the **correct** option.
(1) a and c
(2) Only a
(3) Only c
(4) c and d

174. A probe is generally used in which stage of genetic engineering?
(1) Cleaving DNA and introduction of the identified DNA into the host
(2) Screening, identification of the DNA with desirable genes
(3) Ligation of DNA with vector
(4) Cloning of recombinant DNA in a host

175. Consider the following w.r.t. the reasons of repeated transfusion of genetically engineered lymphocytes required in patients of ADA deficiency
a. The introduced gene gets mutated within few days.
b. Life span of transfused lymphocytes is limited.
c. Enzyme required gets degraded after 20 days of transfusion.
d. Transfused cells get phagocytosed in the host body.
Select the option containing **correct** reason(s) only.
(1) a and b
(2) b and d
(3) a, b and c
(4) b only

176. Read the following statements and choose the **correct** option.
Statement A : The choice of Bt toxin genes depends on the crop and the targeted pest.
Statement B : *B. thuringiensis* forms protein crystals throughout their growth phase.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both A and B statements are correct
(4) Both A and B statements are incorrect

177. In our body, insulin is synthesized within β-cells of ___A___ in inactive form called ___B___, which is activated during its maturation by removal of ___C___.
Choose the option which fill the blanks A, B and C correctly.

	A	B	C
(1)	Pancreatic acini	Insulin	A-peptide
(2)	Pancreas	Proinsulin	C-peptide
(3)	Islets of Langerhans	Proinsulin	A-peptide
(4)	Liver	Proinsulin	B-peptide

(1) (1)
(2) (2)
(3) (3)
(4) (4)

178. Match column I with column II and choose the **correct** option.

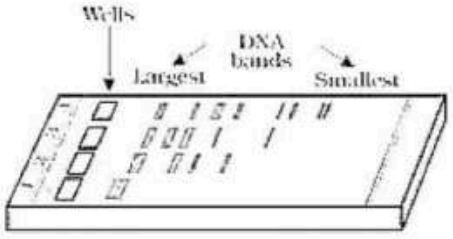
Column I	Column II
a. *crylAc*	(i) Cotton bollworms
b. *crylIAb*	(ii) Corn borers
c. *crylAb*	

(1) a(i), b(ii), c(i)
(2) a(ii), b(i), c(ii)
(3) a(i), b(i), c(ii)
(4) a(ii), b(ii), c(i)

179. 'Presence of insert results into insertional inactivation of the β-galactosidase coding gene and the bacterial colonies do not produce any colour in the presence of a chromogenic substrate'.
If in the above given scenario, insertional inactivation does not occur, then which coloured bacterial colonies will be produced?
(1) White
(2) Blue
(3) Red
(4) Green

180. Which of the following is not a true statement?
(1) Disarmed Ti plasmid is pathogenic to plants.
(2) Ti plasmid can deliver the foreign DNA into the host.
(3) Ti plasmid can act as a vector in its modified form for dicot plants.
(4) Original Ti plasmid is harboured by *Agrobacterium tumefaciens*.

181. Choose the correct option w.r.t. the figure given below.



(1) Lane 3 shows undigested DNA fragments
(2) The matrix used is a synthetic polymer prepared from sea weeds
(3) Smallest band is towards cathode
(4) Bands are visualised after staining it with ethidium bromide followed by exposure to UV light

182. 95% of all the existing transgenic animals are
(1) Monkey
(2) Mice
(3) Cow
(4) Sheep

183. **Assertion (A):** Genetic engineering is a laboratory process by which an individual's genome is deliberately modified.
Reason (R): Organisms whose genes have been altered by manipulation are called GMOs.
In the light of above statements, select the **correct** option.
(1) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion
(2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion
(3) Assertion is true but Reason is false
(4) Both Assertion and Reason are false

184. Read the following statements and choose the option which **correctly** represents True(T) and False(F) statements.
(a) Traditional hybridisation procedures used in plant and animal breeding often lead to inclusion of undesirable genes along with the desired genes.
(b) Biotechnology allows inclusion of only desirable genes into the target organisms.
(c) Bioprocess engineering involves maintenance of microbial contamination-free ambience.
(d) An alien piece of DNA is linked with the origin of replication site so that, it can make multiple identical copies of any template DNA.
(1) (a)-T, (b)-T, (c)-T, (d)-F
(2) (a)-T, (b)-T, (c)-T, (d)-T
(3) (a)-T, (b)-F, (c)-T, (d)-F
(4) (a)-F, (b)-F, (c)-T, (d)-T

185. Read the following statements
Statement-A : Restriction enzymes are obtained only from eukaryotes.
Statement-B : The first discovered restriction endonuclease was *EcoR*I.
Choose the **correct** option.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Statement A is correct but B is incorrect
(4) Statement B is correct but A is incorrect

SECTION-B

186. Read the following carefully.
(a) Vaccine safety
(b) Study of diseases
(c) Biological products
Select the option with the advantages of transgenic animals from above.
(1) (a) and (b) only
(2) (b) and (c) only
(3) (a) and (c) only
(4) (a), (b) and (c)

187. Choose the **incorrect** statement w.r.t. a plasmid.
(1) It is autonomously replicating.
(2) It is an extra-chromosomal DNA.
(3) It is double stranded and circular
(4) It is always single stranded and linear.

188. If a double-stranded linear DNA fragment contains two restriction sites for restriction endonuclease *Pst* I, then how many fragments will be obtained after complete digestion with the same restriction endonuclease?
(1) One
(2) Two
(3) Four
(4) Three

189. Thermostable enzyme *Taq* polymerase is isolated from thermophilic bacteria *Thermus aquaticus*. This enzyme is considered as a
(1) DNA ligase
(2) RNA polymerase
(3) DNA polymerase
(4) Restriction endonuclease

190. There are an estimated _____ varieties of rice in India alone and _____ documented varieties of Basmati are grown in India.
Choose the option that correctly fill the blanks respectively.
(1) 20,000; 27
(2) 2,70,000; 20
(3) 10,000 ; 30
(4) 200,000 ; 27

191. A protoxin is
(1) A primitive toxin
(2) An active toxin
(3) Toxin produced by protozoa
(4) An inactive toxin

192. **Assertion (A) :** Bacterial cells are made competent before transformation.
Reason (R) : DNA is a hydrophilic molecule and cannot pass through cell membrane.
In the light of above statements, select the **correct** option.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

193. How much approximate amount of human α-lactalbumin can be found in 10 litres of milk produced by the first transgenic cow, Rosie?
(1) 2.4 gms
(2) 240 gms
(3) 0.024 kg
(4) 0.24 kg

194. Golden rice is rich source of precursor of
(1) Vitamin D
(2) Vitamin C
(3) Vitamin B
(4) Vitamin A

195. Selectable marker
(1) Helps in selecting transformants from non transformants.
(2) Helps in mixing of non-recombinants and recombinants
(3) Gene is absent in cloning vector pBR322.
(4) Are genes encoding resistance to allergies

196. In order to link the alien DNA, the vector needs to have very few, preferably _____, recognition site(s) for commonly used restriction enzymes.
Choose the **correct** option to fill in the blank.
(1) Four
(2) Three
(3) Two
(4) Single

197. How many of the following given in a box below are required for PCR?

DNA polymerase, pBR322, Primers, Wild *E.coli*, Restriction endonuclease

Select the **correct** option.

(1) One
(2) Two
(3) Three
(4) Four

198. Select the **incorrect** statement.

(1) Proteins which are obtained by recombinant DNA technology are called recombinant proteins.
(2) A stirred-tank reactor is usually cylindrical or with a curved base to facilitate the mixing of the reactor contents
(3) After purification, the product has to be formulated with suitable preservatives .
(4) Downstream processing and quality control testing do not vary from product to product.

199. **Assertion (A):** Insulin cannot be orally administered to diabetic patient.
Reason (R): Insulin is a protein hormone and it is broken down with the action of pepsin in stomach.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A).
(3) (A) is true , (R) is false
(4) (A) is false, (R) is true

200. Select the option which represents the **correct** sequence of processes involved in recombinant DNA technology
(a) Isolation of DNA with desirable gene
(b) Introduction of desired DNA fragment into vector
(c) Culturing of host cells in nutrient medium on large scale
(d) Downstream processing
(e) Transferring the rDNA into the host

(1) (a) → (b) → (e) → (c) → (d)
(2) (a) → (b) → (c) → (d) → (e)
(3) (e) → (d) → (c) → (b) → (a)
(4) (a) → (c) → (b) → (e) → (d)